

Small Numbers but Big Bubbles

How the Law of Small Numbers can help explain bubbles

Gabriele Marchini, supervised by Runjie Geng

ICEF, Higher School of Economics

June 9, 2026

The Goal

One goal: the creation of a theoretical model to explain bubbles based on:

- Realistic assumptions (bounded rationality)
- Empirically valid results (Kindleberger phases)
- LSN as psychological foundation of microfoundation (novelty)

Why? - Literature Review

- Rational bubble theory (Blanchard, 1982)
 - Unrealistic conclusions
- Disagreement based (Scheinkman and Xiong, 2003)
 - No explanation about the source of disagreement
- Behavioural, wavering (Barberis et al., 2018)
 - Wavering is too unrealistic of an explanation
- Behavioural, realization utility (Liao et al., 2021)
 - Not mathematically solveable
- Pure mathematical models (Doblas-Madrid, 2012)
 - No endogeneity, path shape problems

Why? - The need for bubble theory

- Inefficient allocation of resources (Economic problem)
- Redistribution of wealth (Social, Economic and Political problem)
- Possible permanent shocks (Economic problems)

What is a bubble though? Following Brunnermaier (2001)

$$B_t = P_t - F_t$$

In my paper, to distinguish it from "pure" mispricing:

$$|B_t| > \bar{B}$$

The Model - Features

- Discrete time ($t \in I$)
- 2 assets, 1 risky 1 risk-free (0 return)
- Walrasian equilibrium (0 excess demand)
- Complete and perfect market
- A big increase in fundamentals causes the bubble

My main contribution: A mathematical representation of LSN following Rabin (2002)

Agents and market - Features and properties

Agents - Features:

- 1 CARA utility function
- 2 Normal distribution of beliefs

Market - Properties:

- Liquidity traders (trade randomly)
- Fundamentalist (price distributed around fundamental)
- LSN investors (Urn like beliefs)
 - Momentum (No initial info)
 - Reversal (Initial belief)
- Each category in fixed proportions

Liquidity investor

This type of investor trades randomly and unpredictably based on liquidity needs. Its demand is simply given by:

$$X_t^{L*} = \varepsilon_t$$

where

$$\mathbb{E}(\varepsilon_t) = 0, \mathbb{E}(\varepsilon_t^2) = \sigma_\varepsilon^2$$

Fundamentalist - Proposition 1

This type of investor believes that price is normally distributed around the fundamental value F_t .

$\tilde{P}_{t+1}^f \sim N(F_t, \sigma_\nu^2)$, the fundamental value follows a random walk $F_t = F_{t-1} + \nu_t, \nu_t \sim N(0, \sigma_\nu^2)$

At each point in time he solves

$$\max_{X_t^f} \int_{-\infty}^{+\infty} e^{-\gamma(w_t + X_f(P_t + \Delta P - P_t))} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu)^2} d\Delta P$$

Its solution is given by $X_t^{f*} = \frac{F_t - P_t}{\gamma \sigma_\nu^2}$

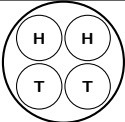
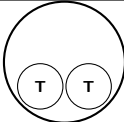
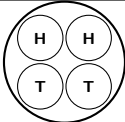
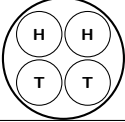
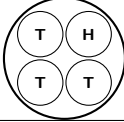
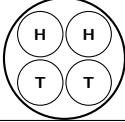
LSN investor (1)

Formal description follows Rabin (2002) with some modifications

- Urn containing N balls (Like Rabin)
- $\lambda_{t-1}N$ positive price increase balls, $(1 - \lambda_{t-1})N$ negative price increase balls (Same as Rabin)
- At each point M large (In Rabin $M=1$), but small compared to N , balls extracted from the urn without replacement
- Urn gets continuously and partially replenished (In Rabin every 2 periods full replenishment)

LSN - Example

Head and Tail. At $i=0$ 2 H balls extracted. At $i=1$ 2 T balls extracted. $i=2$ equilibrium, game starts all over again

	$i = 0$	$i = 1$	$i = 2$
Rabin (2002)			
My modification			

LSN investor (2)

Bernoulli of positive ball, since M small compared to N but M itself large, normal approximation to binomial holds

$$M_t^+ \sim N(\lambda_{t-1}M, M\lambda_{t-1}(1 - \lambda_{t-1}))$$

Then, we construct the believed change in price

$$\Delta P_t = P_{t+1} - P_t = M_t^+ - M^- = 2M_t^+ - M$$

The distribution of price change is thus known:

$$\Delta P \sim N(M(2\lambda_{t-1} - 1), 4M\lambda_{t-1}(1 - \lambda_{t-1}))$$

LSN investor - Proposition 2

Given we found the distribution of the price change we can thus proceed to state the utility maximisation problem

$$\max_{X_t^{sn}} \int_{-\infty}^{+\infty} e^{-\gamma(w_t + X_t^{sn}(P_t + \Delta P - P_t))} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu)^2} d\Delta P$$

Its solution is given by $X_t^{sn*} = \frac{2\lambda_{t-1}-1}{\gamma 4\lambda_{t-1}(1-\lambda_{t-1})}$

LSN investor - The contrarian (1)

Given the LSN, what distinguishes contrarian and momentum is having initial beliefs about lambda or estimating them.

The contrarian is the agent that starts with **certain** beliefs about lambda. Consider price change:

$$\Delta P = P_{t+1} - P_t = 2M_t^+ - M \Rightarrow M_t^+ = \frac{M + \Delta P_t}{2}$$

He finds out the number of positive price balls extracted given the price change

LSN investor - The contrarian (2)

Update of the urn is given by:

$$N_{t+1}^+ = N_t^+ - M_t^+ + 0.5M + F(I_t)$$

Dividing by N both sides one gets the following

$$\lambda_{t+1} = \lambda_t - \xi_t^+ + 0.5\xi + \frac{F(I_t)}{N} \Rightarrow \delta\lambda = \frac{\xi - 2\xi_t^+}{2} + \Phi(I_t)$$

where

$$\xi^+ = \frac{M_t^+}{N} = \frac{M + \Delta P}{2N}, \xi = \frac{M}{N}$$

LSN investor - The momentum (1)

The momentum investor starts with **unknown** composition of the urn. Thus, he estimates the value of lambda, the simplest estimation is given by:

$$\mathbb{E}[\mathbb{E}[M_t^+ | M_t^+] = \mathbb{E}[\lambda_{t-1} M | M_t^+]$$

$$\mathbb{E}[\lambda_{t-1} | M_t^+] = \frac{M_t^+}{M} = \xi^{-1} \xi_t^+ = \frac{\Delta P + M}{2M}$$

Thus, the following yields the difference in lambdas by using earliest available data

$$\delta\lambda = \xi^{-1} \delta\xi_{t-2}^+, \mathbb{E}(\lambda_t) = \xi^{-1} \xi_{t-2}^+$$

LSN investor - The momentum (Corollary 2)

The belief structure of the extrapolator in Barberis et al.(2018) can be described as a case of the LSN under the proposed belief structure by using a weighting process for the estimated lambda.

$$\hat{\lambda} = (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \mathbb{E}[\lambda_{t-i}^m | M_{t-i+1}^+]$$

$$(1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \frac{\Delta P_{t-i-1}}{M} = (1 - \theta) \sum_{k=1}^{\infty} \theta^{k-1} (P_{t-k} - P_{t-k-1})$$

Equal for $M=1$ and equality of weights

LSN investor - The momentum (Corollary 3)

The demand function of the momentum investor is equivalent to a restricted version of the order flow of the momentum investor in Harris Hong and Stein (1999) model under $k=1$ when the price changes are small.

$$X_t^m = \frac{M\Delta P_{t-2}}{\gamma(M^2 - \Delta P_{t-2}^2)} = \frac{\Delta P_{t-2}}{\gamma M} + o(\Delta P_{t-2}) \approx \frac{\Delta P_{t-2}}{\gamma M}$$

Equivalent to $F_t = A + \phi\Delta P_{t-2}$ under $\phi = \frac{1}{\gamma M}$

The market - Equilibrium (Proposition 3)

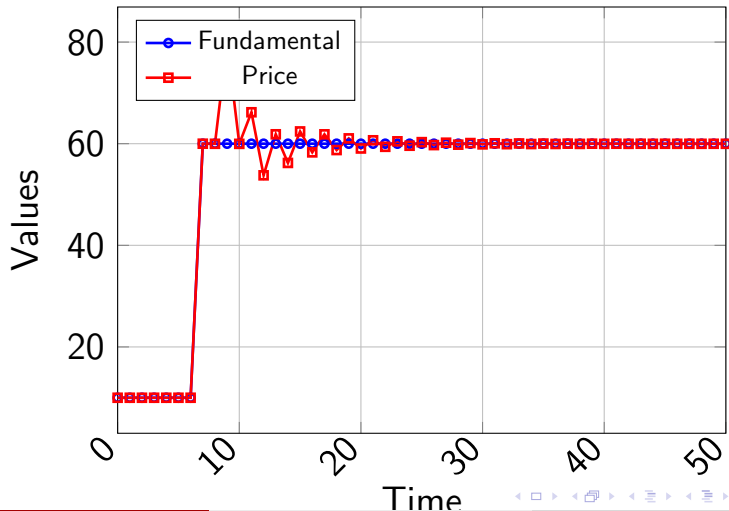
The market clearing price exists and is unique and it is given by:

$$P_t = \frac{\gamma\sigma_\nu^2}{\theta} \left[X_t^L + (1 - \theta) \left(\tau \frac{2\lambda_{t-1}^m - 1}{4\gamma\lambda_{t-1}^m(1 - \lambda_{t-1}^m)} + (1 - \tau) \frac{2\lambda_{t-1}^c - 1}{4\gamma\lambda_{t-1}^c(1 - \lambda_{t-1}^c)} \right) \right] + F_t$$

- Theorem 1: Unique stable steady state at $P=F$ in LR
- Theorem 2: Price is expected to growth at risk free rate in steady state
- Theorem 3: Oscillatory dynamics to reach steady state

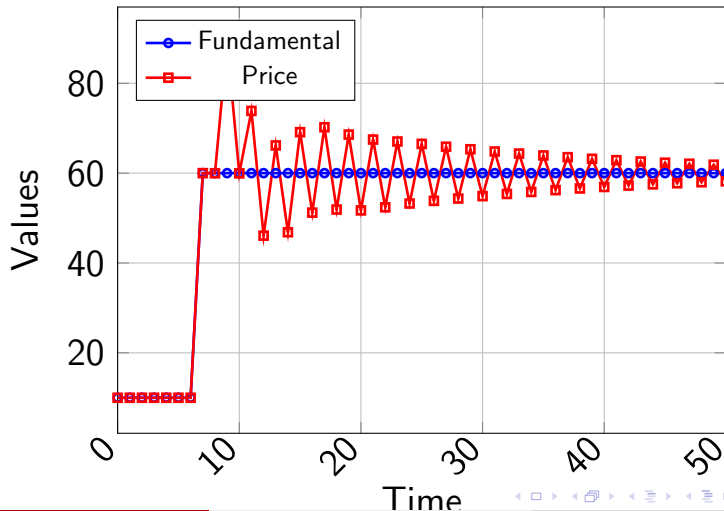
Simulation of price path (Fundamental bubbles)

Fundamental and price over time



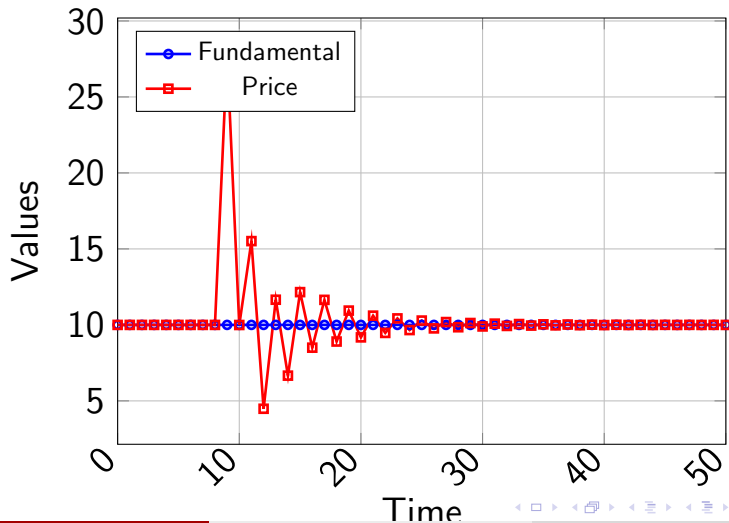
Simulation of price path (Consecutive bubbles)

Fundamental and price over time

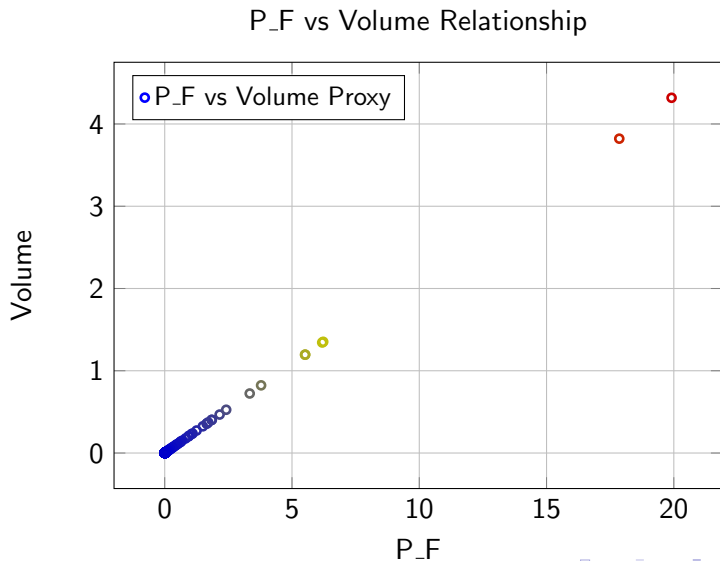


Simulation of price path (Liquidity bubbles)

Fundamental and price over time



Volume in the bubble



Kindleberger phases

- 1 Displacement: At $t=7$
- 2 Expansion: Between $t=7$ and $t=8$
- 3 Euphoria: At $t=8$, $t=9$ and $t=11$
- 4 Crisis: After $t=9$
- 5 Contagion: At $t=10$ and $t=11$

Conclusion

To sum everything up:

- The model is microfounded and follows Rabin (2002) description of the LSN
- The model reflects the Kindleberger phases
- The model allows for different types of crisis but mainly through exogenous events
- Gives psychological foundation to extrapolation
- Around $\Delta P = 0$ momentum investor demands follows Harris Hong et al. (1999) structure
- Bubbles can always be corrected with right liquidity

Appendix - Proof of Proposition 1 (1)

$$\max_{X_t^f} \int_{-\infty}^{+\infty} e^{-\gamma(w_t + X_t^f(P_t + \Delta P - P_t))} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu)^2} d\Delta P$$

Rearranging and taking out of the integral non stochastic parts

$$-e^{-\gamma(w_t + X_t^f\mu + \frac{1}{2}(X_t^f)^2\gamma\sigma^2)}$$

$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}[(\Delta P - \mu)^2 + 2\sigma^2\gamma X_t^f(\Delta P - \mu) + \gamma^2\sigma^4(X_t^f)^2]} d\Delta P$$

Part in the integral is a perfect square

$$(\Delta P - \mu)^2 + 2\sigma^2\gamma X_t^{sn}(\Delta P - \mu) + \gamma^2\sigma^4(X_t^{sn})^2 = (\Delta P - \mu + \sigma^2\gamma X_t^{sn})^2$$

Appendix - Proof of Proposition 1 (2)

Substituting

$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu + \sigma^2\gamma X_t^f)^2} d\Delta P = 1$$

Part to maximise is thus the external exponential

$$\max_{X_t^f} -e^{-\gamma(w_t + X_t^f\mu + \frac{1}{2}(X_t^f)^2\gamma\sigma^2)}$$

After taking first order condition it becomes

$$\mu - X_t^f\gamma\sigma^2 + 0 \Rightarrow X_t^{f*} = \frac{\mu_t}{\gamma\sigma_t^2}$$

Substituting for the mean and variance of the distribution yields

$$X_t^{f*} = \frac{F_t - P_t}{\gamma\sigma_v^2}$$

Appendix - Proof of Proposition 2 (1)

$$\max_{X_t^{sn}} \int_{-\infty}^{+\infty} e^{-\gamma(w_t + X_t^{sn}(P_t + \Delta P - P_t))} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu)^2} d\Delta P$$

Rearranging and taking out of the integral non stochastic parts

$$-e^{-\gamma(w_t + X_t^{sn}\mu + \frac{1}{2}(X_t^{sn})^2\gamma\sigma^2)}$$

$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}[(\Delta P - \mu)^2 + 2\sigma^2\gamma X_t^{sn}(\Delta P - \mu) + \gamma^2\sigma^4(X_t^{sn})^2]} d\Delta P$$

Part in the integral is a perfect square

$$(\Delta P - \mu)^2 + 2\sigma^2\gamma X_t^{sn}(\Delta P - \mu) + \gamma^2\sigma^4(X_t^{sn})^2 = (\Delta P - \mu + \sigma^2\gamma X_t^{sn})^2$$

Appendix - Proof of Proposition 2 (2)

Substituting

$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\Delta P - \mu + \sigma^2\gamma X_t^{sn})^2} d\Delta P = 1$$

Part to maximise is thus the external exponential

$$\max_{X_t^{sn}} -e^{-\gamma(w_t + X_t^{sn}\mu + \frac{1}{2}(X_t^{sn})^2\gamma\sigma^2)}$$

After taking first order condition it becomes

$$\mu - X_t^{sn}\gamma\sigma^2 + 0 \Rightarrow X_t^{sn*} = \frac{\mu_t}{\gamma\sigma_t^2}$$

Substituting for the mean and variance of the distribution yields

$$X_t^{f*} = \frac{F_t - P_t}{\gamma\sigma_\nu^2}$$

Appendix - Proof of Corollary 2 (1)

Let the following be the estimation of lambda for the momentum investor;

$$\hat{\lambda} = (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \mathbb{E}[\lambda_{t-i}^m] = (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \frac{M + \Delta P_{t-i-1}}{2M}$$

The expression can be put as

$$\hat{\lambda} = \frac{M}{2M} + (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \frac{\Delta P_{t-i-1}}{2M}$$

Given the expectation of price difference:

$$\mathbb{E}[\Delta P] = 2\hat{\lambda} - 1 = \frac{M}{M} + (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \frac{\Delta P_{t-i-1}}{M} - 1$$

Appendix - Proof of Corollary 2 (2)

Reconciling it with Barberis (2018)

$$\mathbb{E}[\Delta P] = (1 - \theta) \sum_{k=1}^{\infty} \theta^{k-1} (P_{t-k} - P_{t-k-1})$$

equating the expectations yields the following expression

$$\frac{M}{M} + (1 - \zeta) \sum_{i=1}^{\infty} \zeta^{i-1} \frac{\Delta P_{t-i-1}}{M} - 1 = (1 - \theta) \sum_{k=1}^{\infty} \theta^{k-1} (P_{t-k} - P_{t-k-1})$$

the equation presented above under the restrictions for weights between 0 and 1, of probability and of positivity, holds only for

$$M = 1, N \in I, \zeta = \theta$$

Appendix - Proof of Corollary 3 (1)

Given the following demand for the momentum investor

$$X_t^m = \frac{M\Delta P}{\gamma(M^2 - \Delta P^2)}$$

Compute now a first order Taylor approximation of the expression, it yields for the price difference, it yields the following:

$$X_t^m = \frac{\Delta P}{\gamma M} + o(\Delta P) \approx \frac{\Delta P}{\gamma M}$$

Consider Harris Hong and Stein (1999) order flow under $k=1$

$$F_t = A + \phi\Delta P$$

Appendix - Proof of Corollary 3 (2)

The two are equivalent under the following restrictions

$$\phi = \frac{1}{\gamma M}$$

No restriction for A is needed since by assumption the model is equivalent to Barberis et. (2018) under $Q=0$

Appendix - Proof of Proposition 3 (1)

Starting from the market clearing condition, it holds that:

$$\sum \omega^i X_t^i = Q = 0$$

Substituting the expression becomes:

$$X_t^L + \theta X_t^f + (1 - \theta)\tau X_t^m + (1 - \theta)(1 - \tau)X_t^c = 0$$

By substituting the demands the expression becomes:

$$X_t^L + \theta \frac{F_t - P_t}{\gamma \sigma^2} + (1 - \theta)\tau \frac{2\lambda_{t-1}^m - 1}{4\gamma\lambda_{t-1}^m(1 - \lambda_{t-1}^m)} + \\ (1 - \theta)(1 - \tau) \frac{2\lambda_{t-1}^c - 1}{4\gamma\lambda_{t-1}^c(1 - \lambda_{t-1}^c)} = 0$$

Appendix - Proof of Proposition 3 (2)

As it can be noted, the expression is strictly linear in price, which guarantees uniqueness. Moreover, the equation always has a solution as it is a linear equation. Rearranging the equation yields

$$P_t^* = \frac{\gamma\sigma_\nu^2}{\theta} \left[X^L + (1-\theta) \left(\tau \frac{2\lambda_{t-1}^m - 1}{4\gamma\lambda_{t-1}^m(1 - \lambda_{t-1}^m)} + (1-\tau) \frac{2\lambda_{t-1}^c - 1}{4\gamma\lambda_{t-1}^c(1 - \lambda_{t-1}^c)} \right) \right] + F_t$$

Appendix - Proof of Theorem 1 (1)

By definition at steady state being reached at time t for the first time, clearly:

$$P_t^* = P_{t+k}^*, \forall k : k \in I \wedge k > 0$$

By the definition of $\xi^+(t)$ given that $P(t)$ is constant after time t^* in steady state, it holds that $\Delta P = 0$ and thus $\xi_t^+ = \frac{\Delta P + M}{2N} = \frac{\Delta P}{2N} + \frac{\xi}{2} = \frac{\xi}{2}$ in steady state. By the evolution of lambdas in steady state:

$$\delta\lambda^c = \frac{\xi - 2\xi_t^+}{2} = 0$$

$$\delta\lambda^m = \frac{N\delta\xi_t^+}{M} = 0$$

Appendix - Proof of Theorem 1 (2)

If the following holds it must be that

$$\lambda_t^c = \lambda_{t^*}^c + \sum_{t^*}^t \delta \lambda^c = 0.5$$

$$\lambda_t^m = \frac{M^+}{N} = \frac{N}{M} \xi_t^+ = \xi^{-1} \frac{\xi_t^+}{2} = 0.5$$

Thus,

$$X_t^c = \frac{2\lambda_{t-1}^c - 1}{4\gamma\lambda_{t-1}^c(1 - \lambda_{t-1}^c)} = 0$$

$$X_m = \frac{2\lambda_{t-1}^m - 1}{4\gamma\lambda_{t-1}^m(1 - \lambda_{t-1}^m)} = 0$$

$$X_t^c, X_t^m = 0 \Rightarrow \sum \omega^i X_t^i = \omega^f X_t^f = 0$$

Appendix - Proof of Theorem 1 (3)

The expression above holds only for $P=F$, thus, it must be that in steady state $P=F$ and that the price dynamics are not in steady state if this condition is not satisfied.

Appendix - Proof of Theorem 2 (1)

Let the price dynamics be in the unique stable steady state. The following always holds:

$$\mathbb{E}[\Delta P] = \mathbb{E}[\Delta F_t] + \frac{\gamma\sigma^2}{\theta} \mathbb{E}[\Delta X^L | X^L=0] + (1-\theta) \frac{\gamma\sigma_L^2}{\theta} \mathbb{E}[\tau \Delta X_t^c + (1-\tau) \Delta X_t^m]$$

where

$$X_t^c = \frac{2\lambda_{t-1}^c - 1}{4\gamma\lambda_{t-1}^c(1 - \lambda_{t-1}^c)}, X_t^m = \frac{2\lambda_{t-1}^m - 1}{4\gamma\lambda_{t-1}^m(1 - \lambda_{t-1}^m)}$$

F is an external unknown process and that has expectation equal to today's value. In other words, no change is expected. The steady state has by definition no liquidity demand. Given that the process is normal with mean 0, at 0 no change is expected. Moreover, from Theorem 1 it holds that $\delta\lambda = 0$ in steady state and all components expect lambdas are constant in the above equation, but in steady state it must hold that λ is constant as well and equals 0.5.

Appendix - Proof of Theorem 2 (2)

Thus, under the considered normalisation to 0 of the interest rate:

$$\mathbb{E}[\Delta P] = 0 + 0 = (1 + r)P - P = P - P = 0$$

And the price grows at the risk free interest rate.

Appendix - Proof of Theorem 3 (1)

Under Assumption 1 and Assumption 2 at $t=T$ $P=F$ by Theorem 4.
At time $T-1$ the following thus happens, $\Delta P = \Delta F$

$$\delta\lambda_T^c = \frac{\xi - 2(\frac{M+\Delta P}{2N})}{2} = \frac{\xi}{2} - \frac{M + \Delta P}{2N} = -\frac{\Delta P}{2N} = -\frac{\Delta F}{2N}$$

$$\delta\lambda_T^m = \frac{N}{M}\delta\xi^+ = \frac{N}{M}\frac{M + \Delta P}{2N} - \frac{1}{2} = \frac{\Delta P}{2M} = \frac{\Delta F}{2M}$$

Note that given the simulation settings since M is far smaller than N $|\delta\lambda_T^m| > |\delta\lambda_T^c|$, thus, since they both started from the steady state level of 0.5, there is going to be positive excess demand coming from LSN. The price must then increase over the fundamental at $T+2$

$$\Delta P_T = \frac{\gamma\sigma^2}{\theta}[(1-\theta)(\tau\Delta X_T^m + (1-\tau)\Delta X_T^c)]$$

Appendix - Proof of Theorem 3 (2)

Given the value of lambda before were 0.5 and substituting the expression for lambda differences:

$$\Delta P_{T+1} = \frac{2\sigma^2}{\theta} [(1 - \theta)(\tau \frac{\Delta F}{2M} - (1 - \tau) \frac{\Delta F}{2N})]$$

As long as M is smaller than N and M is big enough it holds that $\Delta P_T = a\Delta F$ with $0 < a < 1$. At next time moment, given the increase in price is smaller than previous one, lambda of momentum investor decreases along with the one of contrarian, driving the price down necessarily.

Convexity of demand function for LSN investor over lambda and linearity of fundamentalist one ensure that the equilibrium is reached through smaller and smaller oscillations.

The same is held under the alternative assumptions using liquidity demand.